

● HARD

● ADVANCED MATH

Advanced Math

05

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Which expression is equivalent to $\frac{y+12}{x-8} + \frac{yx-8}{x^2y-8xy}$?

A. $\frac{xy+y+4}{x^3y-16x^2y+64xy}$

B. $\frac{xy+9y+12}{x^2y-8xy+x-8}$

C. $\frac{xy^2+13xy-8y}{x^2y-8xy}$

D. $\frac{xy^2+13xy-8y}{x^3y-16x^2y+64xy}$

A rectangle has an area of 155 square inches. The length of the rectangle is 4 inches less than 7 times the width of the rectangle. What is the width of the rectangle, in inches?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

ADVANCED MATH

$$\frac{12}{n} - \frac{2}{t} = -\frac{2}{w}$$

The given equation relates the variables n , t , and w , where $n > 0$, $t > 0$, and $w > t$. Which expression is equivalent to n ?

- A. $12tw$
- B. $6t - w$
- C. $\frac{w - t}{6tw}$
- D. $\frac{6tw}{w - t}$

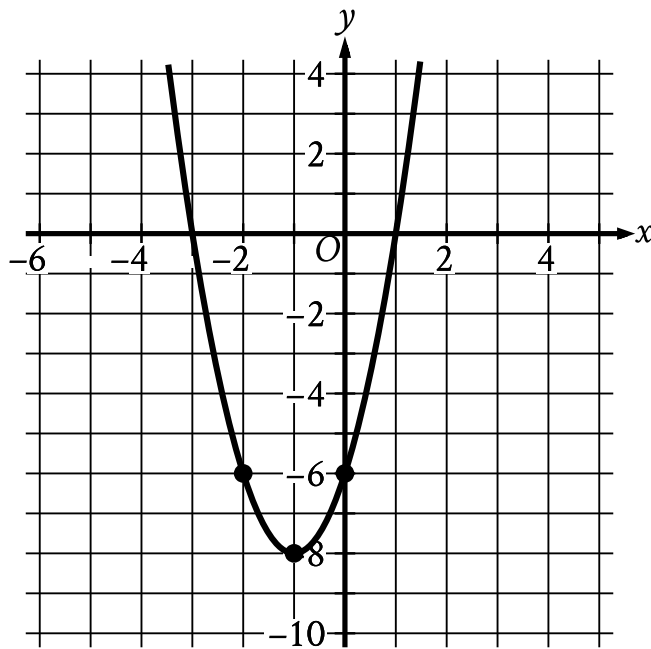
Q5

ADVANCED MATH

The equation $\frac{x^2 + 6x - 7}{x + 7} = ax + d$ is true for all $x \neq -7$, where a and d are integers.

What is the value of $a + d$?

- A. -6
- B. -1
- C. 0
- D. 1



- The parabola opens upward.
- The vertex is at point (negative 1 comma negative 8).
- The parabola passes through the following points:
 - (negative 2 comma negative 6)
 - (negative 1 comma negative 8)
 - (0 comma negative 6)

The graph of $y = 2x^2 + bx + c$ is shown, where b and c are constants. What is the value of bc ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

ADVANCED MATH

$$5x^2 + 10x + 16 = 0$$

How many distinct real solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

The expression $\frac{1}{3}x^2 - 2$ can be rewritten as $\frac{1}{3}(x - k)(x + k)$, where k is a positive constant. What is the value of k ?

- A. 2
- B. 6
- C. $\sqrt{2}$
- D. $\sqrt{6}$

The functions g and h are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $gx = 181.161.4^{x+2}$

II. $hx = 181.4^{x+4}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

$$-5x + 13 = 73$$

What is the sum of the solutions to the given equation?

A. $-\frac{146}{5}$

B. -12

C. 0

D. $\frac{26}{5}$